

~~(2019)~~~~(i)~~

Pg # 221

Olive oil: Extracted from the fruit of olive tree, rich in monosaturated fats & antioxidants, often used in mediterranean cuisine.

Canola oil: Derived from rapeseed, low in saturated fats, & high mono-saturated fats, making it a heart-healthy option for cooking & frying.

Soyabean oil: Extracted from soybeans.

Sunflower oil: Extracted from sunflower seeds.

~~(iii)~~

Importance of Singeing of cotton:

Improved Smoothness:

Singeing gives a fabric

Smooth surface.

Reduced pilling:

By removing loose fibers, singeing reduced pilling and fuzzing.

Enhanced dyeability: Singeing helps dyes penetrate the fabric more evenly.

Improved appearance: Singeing enhances the fabric's appearance by removing unsightly fuzz and fibers.

~~→ (V) →~~

Role of Catechol and pyrogallols in Vegetable tanning?

Catechol and pyrogallols are types of Phenolic Compounds found in plant source. In vegetable tanning they play the following roles.

Catechol: (a type of catechol tannin)

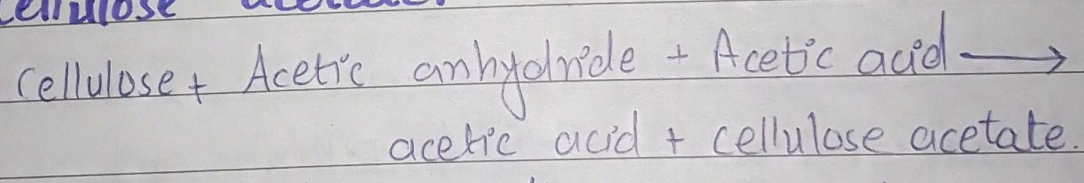
Helps in cross-linkage fibers, increasing leather's strength and water resistance.

Pyrogallol (a type of gallo tannin)

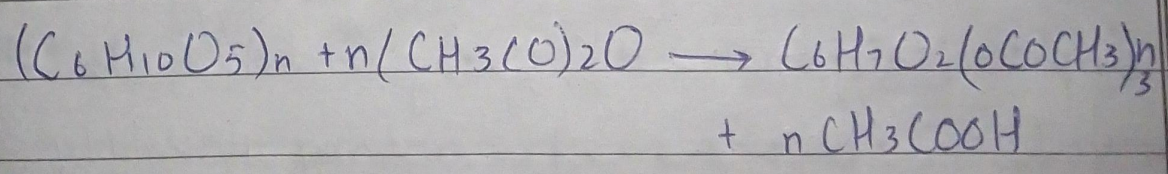
Enhance the softness, color and flexibility of the leather.

~~→ (Vii) →~~

Chemical reaction in manufacturing of Cellulose acetate:



Reaction occurs in the presence of H_2SO_4 as a catalyst.



~~(VIII)~~

Role of liming in Chrome tanning:

Liming in Chrome tanning serve to:

- Remove hair, fats and epidermis from hides.
- Swell and Open up the Collagen fibers for better penetration of tanning agents.
- Break down Keratin to facilitate unhairing.
- Raise the PH to prepare the hide for tanning.

~~(ix)~~

need of Salting in manufacturing of leather?

Preservation: Salt helps to prevent bacterial growth and decay by drawing out moisture from the hides.

Inhibition of microbial growth: Salts creates an environment that inhibits the growth of microorganisms, allowing for longer storage & transportation times.

Stabilization: Salting helps to stabilize the hides, making them easier to handle and process.

~~dx~~

Fibre: (Book pg # 437)

A "Fibre" is defined as any product capable of being woven or otherwise made into fabric. OR

"A 'pliable' hair like strand that is very small in diameter in relation to its length".

Attributes of Fibers:

length: Can be short & long

Strength: Vary in tensile strength.

Elastic: Can be elastic or inelastic

Texture: Smooth, rough or have unique texture.

Absorbency: diff in moisture absorption capacity.

Color: Can be natural or dyed.

Durability: vary in resistance to wear, tear & degradation.

~~(2023)~~

~~(1)~~

Chemical from Ethylene:

Book pg # 318, 319

~~(iii)~~

Nitration of Propane:

Repeat # 2020

Q 2)

Diff problems & Disease Caused by tanning industries to human.

The tanning industry which processes animal hides into leather, poses various health risks to workers & environmental concerns.

- Skin disease
- Respiratory problems (Asthma, bronchitis)
- Cancer
- DNA Mutations
- Neurological damage
- water & soil pollution.

Q VII)

Name "BEAMHOUSE OPERATION" involved in manufacturing of leather. (Book Pg #294)

It is a series of process involved in the initial stages of leather manufacturing. It involves

Soaking: Rehydrating the hide or skin.

Liming: Treating hides with lime to remove fats and hair.

Deliming: Removes excess lime from hide.

Bating: Treating hide with enzyme to soft & relax the fibre.

~~(Xiii)~~

Production of Ethylene from Naphtha:

The production of ethylene from Naphtha involves a process called Steam Cracking.

→ Naphtha is heated with steam in a Cracking Furnace (around 800-900°C) break down into smaller molecules. The resulting gas is cooled separated and purified to produce ethylene through distillation.

~~(x)~~

Essential Oils

- are highly concentrated plant extracts that capture the natural aroma and properties of plant.
- Used in aromatherapy, skin care, and perfumery.
- Are concentrated extracts
- Ex: lavender, tea-tree, Peppermint etc.

Vegetable Oils

are plant based oil extracted from seeds, nuts or fruit.

Used in cooking and food preparation.

Are triglyceride based.

Ex:

olive oil, Coconut oil, Sunflower oil etc.